

Real-Time **ASL** Gesture Recognition

CS 4391 - Final Project

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The Problem

- **Communication Gap:** A massive barrier exists between the deaf/hard-of-hearing community and the general population.
- **Vision Bottleneck:** Standard deep learning (CNNs/ViTs) translates well but is too heavy for portable edge devices.
- **Our Goal:** Build an intuitive, real-time ASL translation system that has high accuracy, operates with low latency, and runs on an edge computing platform.



System Architecture



1. Data

Bootstrapped from Hugging Face & open-source sets. Augmented with custom camera data for sensor-specific fine-tuning.



2. Extraction

MediaPipe Hand Landmarker maps 21 3D joints, generating a normalized 93-dimensional geometric feature vector.



3. Ensemble ML

A custom Scikit-Learn ensemble combining a Random Forest Classifier and an MLP via mathematical soft voting.



4. Edge Inference

Exported to ONNX and deployed on the Jetson Orin Nano via TensorRT execution provider for fast FP16 inference.

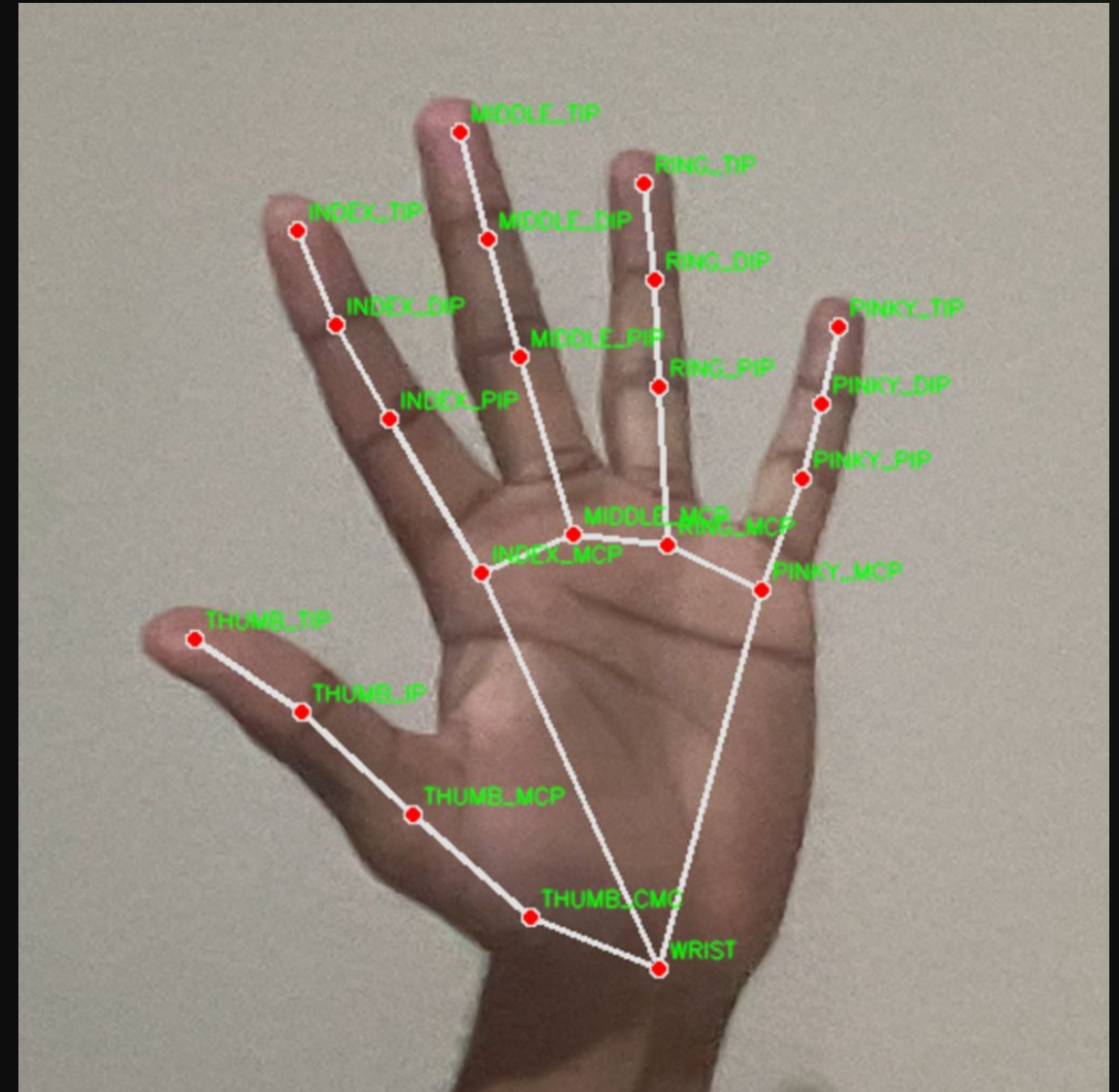
Pose Extraction

Abstracting the Pixel Data

Instead of processing heavy RGB matrices, we pass frames through MediaPipe to extract skeletal geometry.

By flattening 21 structural landmarks into a **93-D feature vector**, we abstract away background noise, skin tone differences, and lighting variations.

This massive dimensionality reduction allows classical ML to outperform heavy deep learning networks on edge hardware.



Optimization & Deployment

Dual Model Ensemble

Our classifier uses two distinct architectures to maximize geometric robustness:

- **Random Forest:** Excels at non-linear boundaries.
- **MLP:** Learns hierarchical joint representations.

Predictions are resolved using a custom soft voting mechanism: $0.5 \times (Prfc + Pmlp)$.

TensorRT Acceleration

Standard Scikit-Learn backends bottleneck CPU resources. We bypassed this by:

- Exporting models to static **ONNX** graphs.
- Executing via the **TensorRT** provider on the Jetson Orin Nano's GPU.

This FP16 optimization unlocked real-time framerates.

Model Performance

96.9%

Overall Test
Accuracy

Robust Results

- **26 Distinct Classes:** Evaluates letters A-Z
- **High Precision:** Majority of letters recognized with a confidence level of at least 90% on the held-out test set.
- **Data Scale:** 54,340 total training samples.
- **Domain Adaptation:** Fine-tuning with heavily weighted custom camera data minimized domain gap and improved real-world reliability.

Live Demo

Thank You!

Questions?
